

Homework 37: Constrained Optimization, Quadratic Forms and Level Curves

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Definition 1. Level Curves and Quadratic Forms. A useful way of visualizing the behavior of a function $f(x, y)$ of two variables is to consider the curves in the xy –plane along which $f(x, y)$ is constant. These curves have equations of the form

$$f(x, y) = k,$$

and are called the level curves of f . In particular, the level curves of a quadratic form $\mathbf{x}^T A \mathbf{x}$ on $\mathbb{R}^2$ have equations of the form

$$\mathbf{x}^T A \mathbf{x} = k,$$

so, the maximum and minimum values of $\mathbf{x}^T A \mathbf{x}$ subject to the constraint $\|\mathbf{x}\| = 1$ are the largest and smallest values of k for which the graph of $\mathbf{x}^T A \mathbf{x} = k$ intersects the unit circle. Typically, such values of k produce level curves that just touch the unit circle, and the coordinates of the points where the level curves just touch produce the vectors that maximize or minimize $\mathbf{x}^T A \mathbf{x}$ subject to the constraint $\|\mathbf{x}\| = 1$.

Problem 1. Revisiting Multivariable Calculus. Find and sketch the level curves $f(x, y) = k$ on the same set of coordinate axes for the given values of k . We refer to these level curves as a contour map.

$$f(x, y) = 5x^2 + 5y^2 + 4xy.$$

Plot the contour map and the unit circle to identify the relative extrema.

Problem 2. Removing Cross-Product Terms. Remove the cross-product term from the following quadratic equation

$$\frac{3}{2}x^2 + \frac{3}{2}y^2 - xy = 4.$$

Problem 3. Indefiniteness and Hyperboloids. Find and sketch the level curves $f(x, y) = k$ on the same set of coordinate axes for the given values of k . We refer to these level curves as a contour map.

$$f(x, y) = xy.$$

Problem 4. Definiteness, Ellipsoids, and Paraboloids. Find and sketch the level curves $f(x, y) = k$ on the same set of coordinate axes for the given values of k . We refer to these level curves as a contour map.

$$f(x, y) = \left(\frac{3}{2}\right)x^2 + xy + \left(\frac{3}{2}\right)y^2.$$

Problem 5. Conceptual Understanding.

i. Show that $Q^T(\mathbf{x}) = Q(\mathbf{x})$.

ii. Show that $Q(k\mathbf{x}) = Q(\mathbf{x})$, for $k \neq 0$, where

$$Q(k\mathbf{x}) = \frac{\mathbf{x}^T A \mathbf{x}}{\mathbf{x}^T \mathbf{x}}.$$

Problem 6. Finding Constrained Extrema. Find the maximum and minimum values of the quadratic form

$$z = 5x^2 + 5y^2 + 4xy,$$

subject to the constraint $x^2 + y^2 = 1$. Study this problem by sketching level curves along with the constraint.

Problem 7. A Constrained Extremum Problem. A rectangle is to be inscribed in the ellipse $4x^2 + 9y^2 = 36$. Use eigenvalue methods to find nonnegative values of x and y that produce the inscribed rectangle with maximum area.

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Definition. Principal Axes. Consider a quadratic form $Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$ where A is a symmetric $n \times n$ matrix with n distinct eigenvalues. Then the eigenspaces of A are called the principal axes of Q . (Note that these eigenspaces will be one-dimensional.)

Let's return to the case of a quadratic form of two variables. We can summarize our findings as follows:

Theorem 1. The Principal Axes Theorem or Diagonalizing a Quadratic Form. Let A be an $n \times n$ symmetric matrix. Then there is an *orthogonal change of variable*, $\mathbf{x} = P\mathbf{y}$, that transforms the quadratic form $\mathbf{x}^T A \mathbf{x}$ into a quadratic form $\mathbf{y}^T D \mathbf{y}$ with no cross-product term. That is

$$Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x} = \mathbf{y}^T D \mathbf{y} = \lambda_1 y_1^2 + \lambda_2 y_2^2 + \cdots + \lambda_n y_n^2.$$

Theorem 2. Ellipses and Hyperbolas. Consider the curve C in $\mathbb{R}^2$ defined by

$$Q(x_1, x_2) = ax_1^2 + bx_1x_2 + cx_2^2 = 1.$$

Let λ_1 and λ_2 be the eigenvalues of the matrix

$$\begin{bmatrix} a & b/2 \\ b/2 & c \end{bmatrix}$$

of Q . If both λ_1 and λ_2 are positive, then C is an ellipse. If one eigenvalue is positive and the other is negative, then C is a hyperbola.

Good Luck